# UAFDcd: Uncertainty-Aware Feature Decoupling Network for Remote Sensing Change Detection

Bin Wang, Sichao Qiao, Ziyv Zhao, Pinle Qin, and Jianchao Zeng

***Abstract*—Remote sensing change detection (RSCD) aims to identify regions in co-registered bi-temporal images that undergo specific changes over time. However, most existing deep networks for RSCD typically adopt mixed feature extraction and fusion. From a causal perspective, this is detrimental to clear class boundary delineation and thus exerts a negative impact on model performance. To address this issue, this article proposes an uncertainty-aware cross-domain feature decoupling network for RSCD. Firstly, this article systematically verifies the rationality of feature disentanglement for RSCD based on causal inference theory. The feature decoupler is constructed through correlation analysis to extract shared features of bi-temporal images, and differential disentanglement is used to separate unique features and obtain change results, which effectively cuts off the causal backdoor path from invariant features to CD outputs. Then, decoupled prototype contrastive learning is introduced to enhance the discriminability of decoupled features. In the specific decoupled feature space, corresponding pixels are pulled close to positive prototypes and pushed away from negative prototypes, while low attention weights are assigned to heterogeneous pixels. This not only improves intra-class compactness and inter-class separability, but also further blocks the backdoor paths of confounding factors affecting prediction results. Finally, an uncertainty estimation module is introduced to improve the confidence of decoupled features. This module learns change and invariant evidence from decoupled features and constructs their parameterized Dirichlet distribution. It not only produces the probabilities of change and invariance but also quantifies model uncertainty and reduces the model epistemic uncertainty via evidential loss. Experiments demonstrate that the proposed method achieves significant improvements in both qualitative and quantitative metrics.**

***Index Terms*—Decoupled contrastive learning, Dirichlet distribution, feature decoupling, remote sensing change detection (RSCD).**

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## I. Introduction

REMOTE sensing change detection (RSCD) aims to identify and quantify differences between multitemporal RS images of the same geographic area acquired at different times, and to accurately delineate changed regions. It has been increasingly and widely applied in diverse fields, including urban expansion analysis, disaster assessment, military strike monitoring, and vegetation coverage detection [1], [2], [3], [4].

Existing RSCD methods could be roughly categorized into two types: 1) feature extraction-based and 2) domain adaptation-based methods. Specifically, feature-based CD methods can be further divided into traditional and deep learning methods. Traditional CD methods (e.g., image arithmetic, image transformation, and post-classification) often yield suboptimal performance and poor generalization, primarily due to their reliance on handcrafted features. Additionally, some critical hyperparameters (e.g., segmentation thresholds) require professional expertise and extensive practical experience for optimization. Currently, deep learning methods have been widely applied to CD tasks, benefiting from their powerful feature extraction and nonlinear representation. Representative deep networks for RSCD include convolutional neural networks (CNNs), Mamba, denoising diffusion probabilistic models, and their hybrid variants [5], [6], [7].

CNNs extract features by sliding convolution kernels over local regions, enabling them to effectively capture local spatial information. They can efficiently process data with prominent local features (e.g., edges and textures). In addition, deep graph CNNs are also frequently used to encode spatial structural information [8]. Common CNN-based RSCD networks include fully convolutional networks, U-Net, and their variants [9], [10], [11]. Although CNNs can learn correlations between neighboring features, their limited sampling range restricts them from modeling long-range dependencies. To address this limitation, some studies have adopted the multihead self-attention based on Transformers to achieve context modeling in both spatial and temporal domains, thereby effectively improving the global accuracy of RSCD [12], [13]. In addition, the Mamba model integrates a comprehensive selective scanning module that can selectively scan images in multiple directions, enabling it to capture large-scale features from diverse orientations. Mamba has demonstrated strong spatiotemporal relationship modeling and efficient global context capture in RSCD tasks [2], [14], [15]. Qu et al. [16] proposed a spatial-spectral-temporal collaborative Mamba-based active federated hyperspectral CD. Denoising diffusion probabilistic

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models gradually corrupt data structures through forward diffusion and then learn to recover data from noise during reverse denoising, thus supporting tasks, such as data distribution modeling and generation. Recently, researchers have employed multiscale architectures to extract hierarchical features from paired bi-temporal RS images, thereby enhancing model robustness [3], [17]. Although the aforementioned methods have achieved certain results for specific scenarios, RSCD. However, when applied across domains, models struggle to adapt effectively due to discrepancies in data distribution [18], [19]. RSCD methods based on domain adaptation partition bi-temporal images into source and target domains with the goal of enabling models trained on the source domain to generalize effectively to the target domain. For example, researchers have decoupled bi-temporal images into style and content features, directly searching for change patterns in the aligned content feature space to mitigate the adverse effects of style variations (e.g., illumination changes). Feng et al. [20] proposed a cross-domain few-shot learning network for hyperspectral image multiclass CD. Although domain adaptation methods aim to enhance model generalization, current models still exhibit limited generalization capability when confronted with complex and variable RS data. When factors, such as illumination lead to increased distribution divergence in RS data, model performance on the target domain may degrade rapidly.

To date, although researchers have proposed various models to address the challenges in RSCD, most models primarily rely on extracting mixed features to predict changed areas. This reliance gives rise to three core issues: First, distinct feature types (e.g., variable and invariant features) tend to be highly coupled, which hinders the differentiation of independent effects from various factors in feature space and impairs model interpretability. From a causal mechanism perspective, deep features represent high-level yet severely entangled image representations, where all generative factors are conflated and encoded into a single feature space. As shown in Fig. 1, this entanglement significantly degrades the performance of downstream tasks (e.g., segmentation, classification, and detection). Second, there is a lack of explicit modeling for change-related features. Explicit decoupling aims to separate change-specific information (e.g., new buildings and vegetation destruction) from invariant backgrounds. In the absence of such explicit decoupling, existing methods often require secondary conversion of indirect features (e.g., style and content) to implicitly convey change semantics, thereby increasing the complexity of logical reasoning in RSCD. Third, models lack training on epistemic uncertainty. Bi-temporal RS images have inherent uncertainties induced by seasons, illumination, and shadows, etc. That may impede model understanding of change essence and lead to poor generalization.

To address the aforementioned issues, this article proposes UAFDcd based on uncertainty-aware cross-domain feature disentanglement for RSCD. As shown in Fig. 2, the motivation underlying UAFDcd is to directly extract shared public features and unique features from cross-domain bi-temporal images by learning a feature decoupler. Subsequently, the model aims to aggregate intra-class attribute features while enhancing its ability to capture inter-class feature

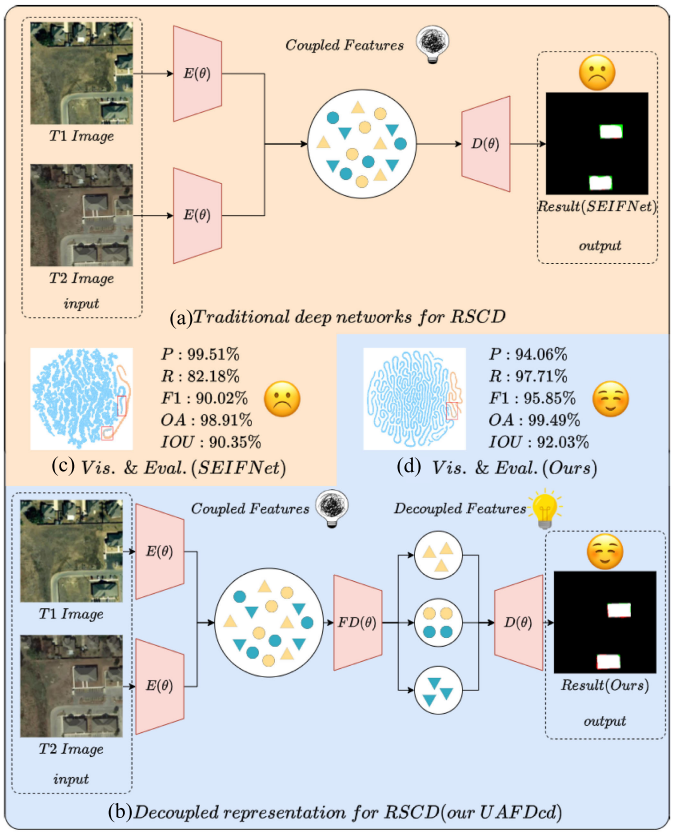



Fig. 1. Illustration of basic architecture on traditional deep learning and our methods. (a) Traditional deep learning-based CD networks that adopt mixed feature extraction and fusion. The internal structure of hybrid features is highly entangled (c), which directly prevents them from distinguishing between "epistemic differences" and "essential changes." This blurs the semantic boundary between changing and unchanging features, thereby adversely affecting CD performance. (b) RSCD networks based on decoupled representations, which divide image-related properties into several relatively independent subsets (i.e., changed and unchanged feature space), whereby CD results exhibit a certain causal relationship with semantic attribute subsets (d).

differences through prototype contrastive learning, thereby enabling extracted features to be more discriminative. Finally, the proposed uncertainty estimation module learns deep evidence corresponding to changed and unchanged classes from the disentangled features and constructs their parameterized Dirichlet distributions. Based on evidential reasoning theory, this module not only outputs the probability values of changed and unchanged classes but also quantifies the model prediction uncertainty. Furthermore, with the assistance of the evidence loss function, it effectively mitigates model epistemic uncertainty and further enhances its generalization performance in complex RS scenarios.

The main contributions are as follows.

1) From the perspective of causal inference, this article systematically demonstrates the idea and implementation of feature decoupling. Specifically, the feature decoupler is designed based on correlation analysis to mine shared features of bi-temporal images; Private features are further obtained via differential decoupling to generate a CD map. This approach can effectively block the causal backdoor paths where invariant features affect change

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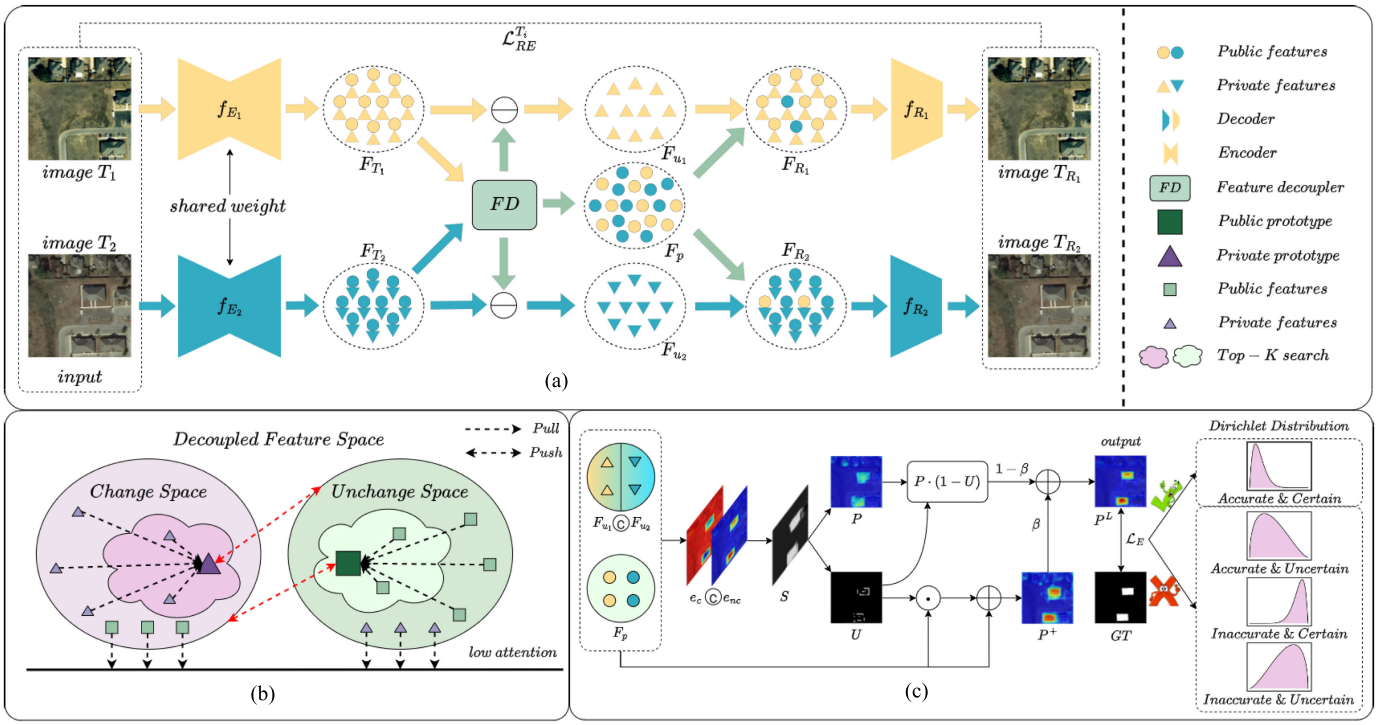



Fig. 2. Pipeline of uncertainty-aware feature decoupling network for RSCD. (a) Feature decoupling and reconstruction. This module achieves feature separation of changed and unchanged information via the feature decoupler (FD), and feature reconstruction is designed to preserve information completeness during the decoupling process. (b) Decoupled contrastive learning. To ensure the intra-class compactness and inter-class separability of decoupled features, the prototype contrastive learning for decoupled features is proposed. (c) Uncertainty estimation module. To reduce model epistemic uncertainty, the CD prediction probability and its uncertainty are modeled as a Dirichlet distribution, aiming to train the high-accuracy and low-uncertainty CD model via the evidence loss.

judgments or changing features lead to predictions of invariant regions, and eliminate interference from irrelevant confounding factors.

2) To further block the backdoor paths of confounding factors affecting prediction results, we enhance the discriminability of decoupled features via prototype contrastive learning. Within the respective decoupled spaces of change and invariant features, corresponding pixels are drawn closer to positive prototypes and distanced from negative prototypes, with low attention assigned to heterogeneous pixels.
3) An evidence-aware uncertainty estimation module is designed to improve the confidence of decoupled features. It first quantifies pixel-wise class probabilities and prediction uncertainty via the Dirichlet distribution, and then enhances the model cognition through evidential loss.

The remainder of this article is organized as follows. Section II briefly reviews related work. Section III elaborates on the model framework and its implementation details. Section IV describes the experimental setup. Section V presents the experimental results and analysis. Section VI addresses the limitations of the method. Section VII concludes this article.

## II. Related Work

### A. Feature Decoupling

Feature decoupling representation aims to separate mixed features into distinct and relatively independent subspaces or representation forms through a specific method or model, where each subspace encodes a specific feature or factor of the data. This approach facilitates the learning of intrinsic data characteristics and enables a clearer understanding of specific features on which the model relies for decision-making [21].

In the field of RSCD, feature decoupling can be roughly categorized into two major types: direct and indirect disentanglement. Indirect disentanglement implicitly characterizes change semantics, and its typical technical pipeline is as follows: it leverages the encoder-decoder network to generate attribute features (e.g., style and content) [22], [23], then conducts land cover difference inference based on specific attribute features like content to achieve RSCD [24]. Feng et al. [25] proposed an adaptive fractional domain decoupling network for hyperspectral CD. However, such methods have obvious limitations. First, during feature encoding and decoding, it is prone to losing fine-grained change information (e.g., building edges). Second, deriving change semantics through indirect attribute features would increase model complexity. In addition, indirect features usually lack available annotation labels, which needs to be addressed through two approaches: either constructing annotated datasets or relying on an unsupervised learning paradigm. By contrast, direct disentanglement takes the core task of RSCD as its intuitive objective. It directly captures land cover change signals through contrastive learning or differential operations, while retaining more differential information in the original images.

### B. Prototype Learning

Prototype learning is to extract representative prototypes from data and use them to represent categories or knowledge,

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thereby enabling tasks such as classification and inference [26], [27]. Prototype learning focuses on the holistic feature representation of a category, which can take the form of a model, a vector, a set of features, etc. For RSCD, input samples can be mapped to the same space as prototypes; subsequently, a similarity metric (e.g., Euclidean distance or Cosine similarity) is defined to measure the proximity between data and prototypes, forming the basis for final decision-making [28]. Since prototypes represent a collective group of samples, downstream tasks based on prototype learning are less susceptible to individual noisy samples, thereby enhancing the model robustness to data noise. Additionally, to accommodate data distribution shifts, some deep learning models have incorporated dynamic prototype update strategies.

### C. Contrastive Learning

Contrastive learning lies in constructing the constraint that governs feature similarities and differences between samples to guide the model to learn highly discriminative feature representations from unlabeled data. This effectively breaks the bottleneck of insufficient labeled data, thereby providing high-quality feature support for downstream tasks, such as classification, detection, and segmentation [29], [30], [31]. From the perspective of learning frameworks, contrastive learning could be categorized into three major types: 1) pairwise contrastive learning; 2) triplet contrastive learning; and 3) negative sample pool-based contrastive learning. Pairwise contrastive learning constructs explicit constraint conditions, which drive the model to shorten feature distance between sample belonged to same class while enlarging those between different-class samples in the feature space, thereby prompting the model to learn highly discriminative feature representations [32]. Lv et al. [33] designed a graph feature learning network to learn the near-far dependencies of graph features for RSCD. Ding et al. [34] proposed a cross-view contrastive loss to enforce structural consistency between views to learn noise-invariant representations. Triplet contrastive learning relies on the triplet network architecture and constructs constraint relationships by introducing three types of samples (i.e., anchor, positive, and negative samples) to boost model capacity for capturing more discriminative features [35], [36], [37]. Contrastive learning based on a negative sample pool enhances the constraint in feature space by constructing and maintaining large-scale negative samples, which effectively improves model representation learning performance while ensuring training efficiency [38], [39], [40], [41].

## III. Method

From the perspective of causal mechanisms, RSCD essentially aims to distinguish the causal factors of real changes from irrelevant interfering factors such as illumination, seasonal variations, noise, etc. Inspired by this, we explicitly decouple features into change-related features and invariant features to construct a more reasonable causal representation.

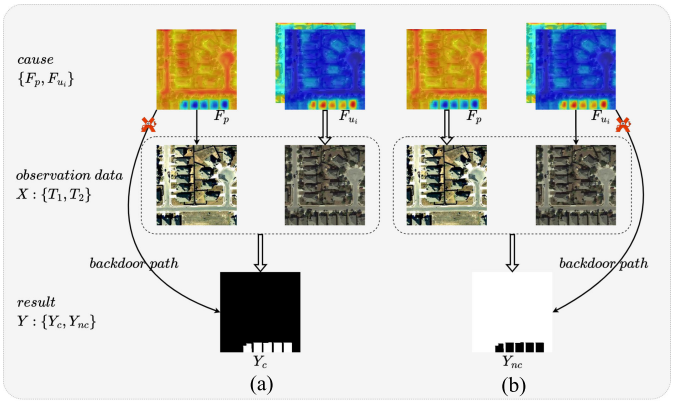



Fig. 3. Structural causal model for RSCD based on feature decoupling. (a) Causal path of changed regions: $F_{u_i} \rightarrow X \rightarrow Y_c$. (b) Causal path of unchanged regions: $F_p \rightarrow X \rightarrow Y_{nc}$.

### A. Structural Causal Model for RSCD

Next, this article elaborates on the relationship between RSCD and causal theory, taking predicted change regions as an example. As shown in Fig. 3(a), we define a directed acyclic graph for the bitemporal RSCD task.

Node Definitions.

1) $F_p$ *(Invariant Feature):* Invariant causal variable (e.g., background information).
2) $F_{u_i}$ *(Change Feature):* Change causal variable (e.g., substantive ground object alteration).
3) $X$ *(Input):* Observed bitemporal image pair $\{T_1, T_2\}$.
4) $Y_c$ *(Outcome):* Change region prediction result.

Edge logic of the causal graph.

1) $F_p \rightarrow X, F_{u_i} \rightarrow X$: The input image is jointly generated by background and change information.
2) $F_{u_i} \rightarrow Y_c$: Causal path—genuine change logically leads to the generation of $Y_c$.
3) $F_p \rightarrow Y_c$: Backdoor path—a spurious correlation path. For instance, the model incorrectly infers a change whenever a "red roof" appears.

Ultimately, the RSCD task based on a structural causal model is transformed into eliminating the interference of backdoor paths via causal intervention $P(Y_c \mid do(X))$.

### B. Feature Disentanglement Based on Causal Inference

This part first conducts the mathematical proof for causal inference of feature decoupling, and then proposes our feature disentanglement method based on that proof conclusion.

As shown in Fig. 2, the bi-phase RS images $T_i \in \mathbb{R}^{4H \times 4W \times 3}$ are processed by feature encoders $f_{E_i}$ to extract deep latent initial features $F_{T_i} \in \mathbb{R}^{C \times H \times W}$. Since feature encoder $f_{E_i}$ incorporates nonlinear activation functions, such as ReLU, the mapping from input image to latent space is highly nonlinear. We define observed image $X$ as a mixture of latent variables $\{F_p, F_{u_i}\}$ generated by an unknown nonlinear function

$$F_{T_i} = f_{E_i}\left(T_p, F_{u_i}\right). \tag{1}$$

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We perform a first-order Taylor expansion of this nonlinear function around the invariant background manifold $(F_p, 0)$

$$F_{T_i} \approx f(F_p, 0) + \left.\frac{\partial f}{\partial F_{u_i}}\right|_{(F_p,0)} F_{u_i} + o\left(\left\|F_{u_i}\right\|\right). \tag{2}$$

For simplicity, we define.

1) $\Phi(F_p) = f(F_p, 0)$: Nonlinear embedding representation of invariant background information $F_p$.
2) $\Delta\Phi(F_{u_i}) = (\partial f/\partial F_{u_i})F_{u_i}$: Local offset vector in feature space induced by change.
3) $o(\|F_{u_i}\|)$ denotes that the remainder is an infinitesimal of higher order with respect to $F_{u_i}$.

At this point, its output in latent space is approximately decomposed into an additive superposition of two independent vectors

$$F_{T_i} \approx \Phi\left(F_p\right) + \Delta\Phi\left(F_{u_i}\right). \tag{3}$$

In terms of a causal graph, we sever the backdoor path $F_p \rightarrow Y_c$ through (4). By explicitly subtracting the contribution of $F_p$ in feature space, the predictor is prevented from "cheating" using background information

$$\begin{aligned} F_{u_i} = F_{T_i} - F_p &\approx \left(\Phi\left(F_p\right) + \Delta\Phi\left(F_{u_i}\right)\right) - \Phi\left(F_p\right) \\ &= \Delta\Phi\left(F_{u_i}\right). \end{aligned} \tag{4}$$

According to the backdoor adjustment in causal inference, we need to perform marginalization by integrating over the invariant background factor $F_p$ to compute the pure causal effect of $X$ on $Y_c$, namely

$$P\left(Y_c \mid do\left(X\right)\right) = \int P\left(Y_c \mid \Delta\Phi\left(F_{u_i}\right), \Phi\left(F_p\right)\right) P\left(F_p\right) dF_p. \tag{5}$$

In decoupled feature space, only the varying local offset vector $\Delta\Phi(F_{u_i})$ is retained, with no dependence on the absolute basis $\Phi(F_p)$. Consequently

$$P\left(Y_c \mid do\left(X\right)\right) \approx P\left(Y_c \mid \Delta\Phi\left(F_{u_i}\right)\right). \tag{6}$$

This mathematically demonstrates that after decoupling, the spurious correlation backdoor path $F_p \rightarrow Y_c$ introduced by invariant background confounding is completely blocked. The model is ultimately constrained to conduct unbiased inference solely through the unique genuine causal path $F_{u_i} \rightarrow X \rightarrow Y_c$.

Next, we will investigate how to extract shared features based on the above proof conclusions.

According to the conditional independence assumption in causal inference: given the background $F_p$, change features are independent (i.e., $F_{u_1} \perp\!\!\!\perp F_{u_2} \mid F_p$).

It follows that:

$$E\left[F_{T_1}^{\top} F_{T_2}\right] \approx E\Big[\left(\Phi\left(F_p\right) + \Delta\Phi\left(F_{u_1}\right)\right)^{\top}\left(\Phi\left(F_p\right) + \Delta\Phi\left(F_{u_2}\right)\right)\Big]. \tag{7}$$

Accordingly, their offset vectors in latent space are mutually independent, and the expectations of cross terms tend to zero

$$E\left[\Phi\left(F_p\right)^{\top}\Delta\Phi\left(F_{u_i}\right)\right] = 0, \quad E\left[\Delta\Phi\left(F_{u_1}\right)^{\top}\Delta\Phi\left(F_{u_2}\right)\right] = 0. \tag{8}$$

Then, the expectation of cross correlation finally collapses to

$$E\left[F_{T_1}^{\top} F_{T_2}\right] \approx E\left[\Phi\left(F_p\right)^{\top}\Phi\left(F_p\right)\right]. \tag{9}$$

Since batchnorm is added after each convolutional layer in the feature encoders $f_{E_i}$, we have $E[F_{T_i}] = 0$. Accordingly, it can be further deduced that $cov[F_{T_1}^{\top} F_{T_2}] = E[F_{T_1}^{\top} F_{T_2}]$. Thus, cross-correlation computation statistically filters out uncorrelated local offset vectors $\Delta\Phi(F_{u_i})$, forcing shared feature $F_p$ to strictly approximate the embedding representation of a nonlinear invariant causal variable.

Inspired by (9), we design a nonlinear feature decoupler to obtain shared public feature representation $F_p$. It consists of two components: one is obtained by measuring the similarity from $\{T_1, T_2\}$ and weighting these average features; the other is learned directly from overall features. As shown in the following equation:

$$\begin{aligned} F_p = &\left(\sigma\left[f_{K_3}^{S_1}\left(F_{T_1}\right) \otimes f_{K_3}^{S_1}\left(F_{T_2}\right)^{\top}\right] / \sqrt{C}\right) \otimes f_{K_3}^{S_1}\left(\frac{F_{T_1} + F_{T_2}}{2}\right) \\ &+ f_{K_3}^{S_1}\left(F_{T_1} \mathbb{C} F_{T_2}\right) \end{aligned} \tag{10}$$

where $f_{k_3}^{s_1}(\cdot)$ represents the convolution operation with kernel size $k = 3 \times 3$ and stride $s = 1$, and $\otimes$ denotes matrix multiplication. $\sigma(\cdot)$ is defined as the $Softmax(\cdot)$ activation function, $\mathbb{C}$ denotes the channel-wise feature concatenation.

To ensure information completeness during feature disentanglement, we employ the image reconstructor $f_{R_i}$ to perform bi-phase image reconstruction based on the shared public features and unique features

$$F_{R_i} = f_{R_i}\left(F_p, F_{u_i}, \theta\right). \tag{11}$$

Subsequently, the reconstruction loss is utilized to ensure the detailed consistency between the original and reconstructed images

$$\mathcal{L}_{RE}^{T_i} = \mathbb{E}_{T_i \sim P(T_i)}\left(\left\|T_{R_i} - T_i\right\|_1\right) \tag{12}$$

where $\|\cdot\|_1$ denotes the $L1$ norm, and $\mathcal{L}_{RE}^{T_i}$ represents the self-reconstruction loss for $T_i$.

### C. Decoupled Contrastive Learning

To further assist the feature decoupler in achieving cross-domain disentanglement representation, the decoupled contrastive learning is specifically designed to pull pixels closer to their corresponding positive prototypes and push them farther away from negative prototypes. As illustrated in Fig. 4, this facilitates the learning of more discriminative features for distinguishing between changed and unchanged pixels.

A prototype can be understood as a feature-level representation of a specific class. Therefore, we first obtain features of changed and unchanged elements by leveraging extracted bi-temporal features and initial prediction probability. The initial change probability map $P^c$ is generated through the medium-term CD head

$$P^c = f_{head}^{mid}\left(F_{T_1} - F_{T_2}, \theta\right). \tag{13}$$

Through thresholding, $P^c$ is converted into a binary map, thereby labeling corresponding positions of input features

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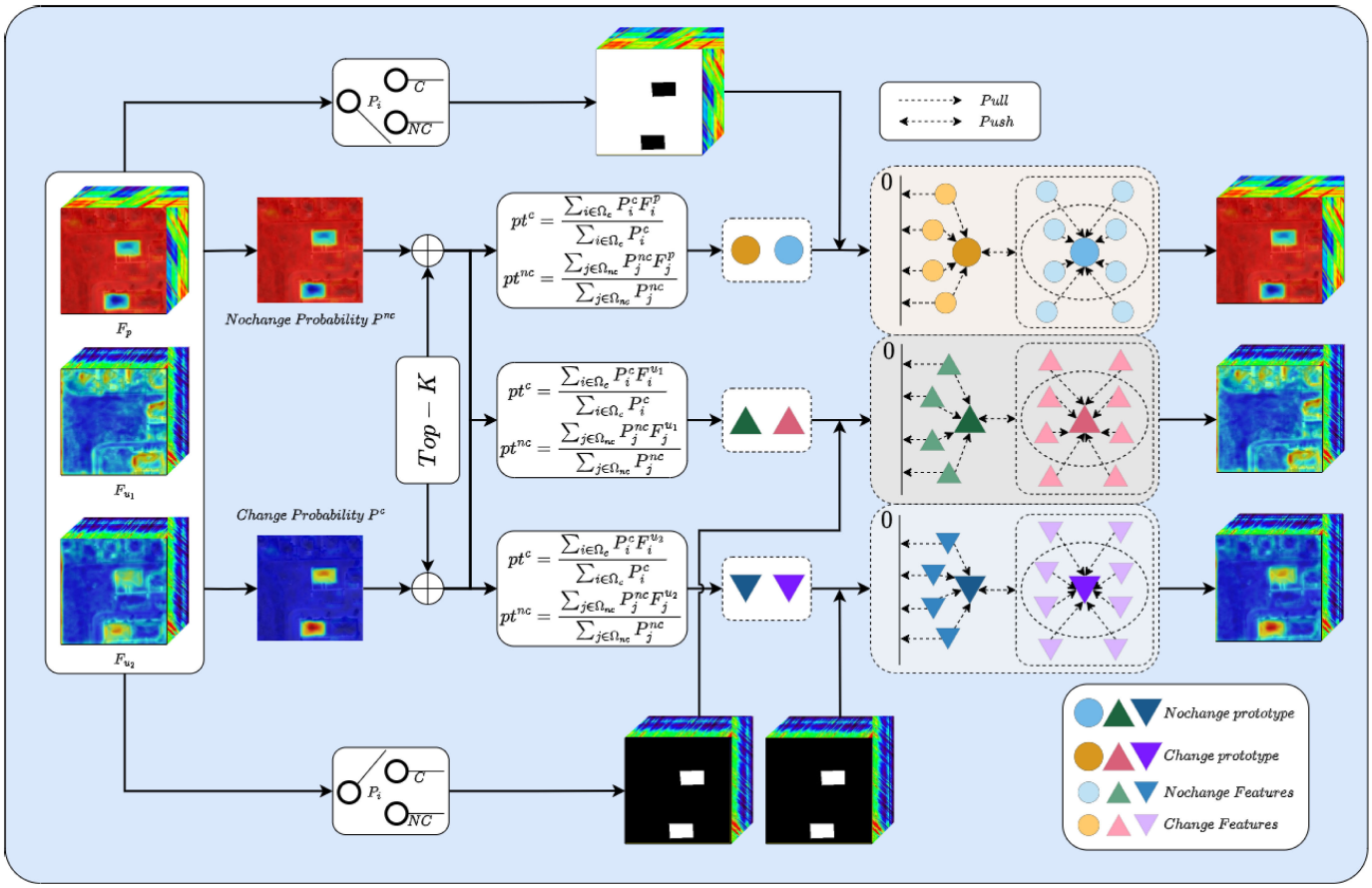



Fig. 4. Pipeline of the decoupled contrastive learning (DCL) mechanism, which depicts prototype generation, the contrastive learning procedure, and final results.

as either changed or unchanged. This result is subsequently utilized for contrastive learning with the prototypes. The probability map $P^{nc}$ for unchanged regions can be obtained by inverting $P^c$, namely

$$P^{nc} = 1 - P^c. \tag{14}$$

Subsequently, the reliable pixel tensor is selected based on the $Top-K$ values, and its corresponding prototype is derived via weighted averaging. The specific steps are as follows:

$$pt^c = \frac{\sum_{i\in\Omega_c} P_i^c F_i}{\sum_{i\in\Omega_c} P_i^c} \tag{15}$$

$$pt^{nc} = \frac{\sum_{j\in\Omega_{nc}} P_j^{nc} F_j}{\sum_{j\in\Omega_{nc}} P_j^{nc}} \tag{16}$$

where $F_i \supseteq [F_p, F_{u_i}]$ and $(P_i^c, P_i^{nc})$ denotes the *ith* element of the initial probability map of $(P^c, P^{nc})$. $(\Omega_c, \Omega_{nc})$ represents the position set corresponding to the $top-K$ values of the changed and unchanged feature tensor. $K$ is a hyperparameter that controls the range of prototype pixels, which is set to 0.1 in this work.

For shared public features, pixels within corresponding regions are expected to be as close as possible to the unchanged prototype, while the attention allocated to changed regions is minimized. Therefore, the prototype loss of shared features $L_{DCL}^p$ is divided into two parts, namely

$$\mathcal{L}_{DCL}^p = \mathcal{L}_c^p + \mathcal{L}_{nc}^p \tag{17}$$

where $\mathcal{L}_{nc}^p$ denotes the prototype loss of unchanged pixels from shared features, and $\mathcal{L}_c^p$ denotes the low-attention loss of changed pixels from shared features

$$\mathcal{L}_{nc}^p = -\frac{1}{N_{nc}} \sum_{i=1}^{N_{nc}} log\left( \frac{e^{\frac{<x_i^{nc}, pt_p^{nc}>}{\tau}}}{e^{\frac{<x_i^{nc}, pt_p^{nc}>}{\tau}} + e^{\frac{<x_i^{nc}, pt_p^{c}>}{\tau}}} \right) \tag{18}$$

$$\mathcal{L}_c^p = \frac{1}{N_c} \sum_{i=1}^{N_c} \left( 1 - e^{-\frac{\|x_i^c, 0\|_2}{\tau}} \right) \tag{19}$$

where $\tau$ represents the temperature coefficient. $N_c$ and $N_{nc}$ denote the number of changed and unchanged pixels, respectively. Additionally, $x_i^c$ and $x_i^{nc}$ refer to the $i$th pixel in the changed and unchanged regions, correspondingly.

For unique features, we expect pixels in changed regions to be close to their respective change prototypes, while the attention weights assigned to unchanged regions are minimized. Therefore, the prototype loss of private features $\mathcal{L}_{DCL}^{u_i}$ is also divided into two parts, namely

$$\mathcal{L}_{DCL}^{u_i} = \mathcal{L}_c^{u_i} + \mathcal{L}_{nc}^{u_i} \tag{20}$$

where $\mathcal{L}_c^{u_i}$ denotes the prototype loss of changed pixels from private features of the $i$th temporal image, and $\mathcal{L}_{nc}^{u_i}$ denotes the

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low-attention loss of unchanged pixels from private features of the $i$th temporal image

$$\mathcal{L}_c^{u_i} = -\frac{1}{N_c}\sum_{j=1}^{N_c} log\left(\frac{e^{\frac{<x_j^c, pt_{u_i}^c>}{\tau}}}{e^{\frac{<x_j^c, pt_{u_i}^c>}{\tau}} + e^{\frac{<x_j^c, pt_{u_i}^{nc}>}{\tau}}}\right) \tag{21}$$

$$\mathcal{L}_{nc}^{u_i} = \frac{1}{N_{nc}}\sum_{j=1}^{N_{nc}}\left(1 - e^{-\frac{\|x_j^{nc}, 0\|_2}{\tau}}\right) \tag{22}$$

where $pt_{u_i}^c$ stands for the change prototype of private feature in $i$th phase image.

In summary, the total loss function of decoupled comparative learning is as follows:

$$\mathcal{L}_{DCL} = \mathcal{L}_{DCL}^p + \sum_{i=1}^{2}\left(\mathcal{L}_{DCL}^{u_i}\right). \tag{23}$$

### D. Uncertainty-Guided Change Decision-Making

Owing to imaging complexity, bi-temporal RS images are often affected by numerous uncertain factors (e.g., varying illumination and floating clouds). Existing RSCD models tend to directly predict the probability of deterministic changes but fail to quantify epistemic uncertainty. Relying solely on the probability of predicted masks may lead to erroneous results. Inspired by Dempster-Shafer evidence theory [42], we propose an evidence-aware uncertainty estimation module. Specifically, this module first collects evidence for changed and unchanged features $(E_c, E_{nc})$ by decoupling representations $(F_{u_i}, F_p)$. It then models pixel-level category probabilities as a Dirichlet distribution and estimates uncertainty U of predicted probabilities. Leveraging this Dirichlet distribution, reliable feature vectors are selected based on subjective logic theory to enhance the model adaptation. For the CD binary classification task, given decoupled features $(F_{u_i}, F_p)$, their evidence vectors $(E_c, E_{nc})$ are derived via the following transformation function:

$$\left[E_c,\ E_{nc}\right] = e^{\frac{\tanh\left[F_{u_1} \copyright F_{u_2}, F_p\right]}{\tau}} \tag{24}$$

where $\tau$ denotes a scale parameter. The confidence scores for changed and unchanged features $(B_c, B_{nc})$ and corresponding uncertainty $U$ are given as follows:

$$\left[B_c, B_{nc}\right] = \frac{\left[E_c, E_{nc}\right]}{S}, \qquad U = \frac{2}{S} \tag{25}$$

where $S = E_c + E_{nc} + 2\cdot\mathbf{1}_{H\times W}$, such that $U + B_c + B_{nc} = \mathbf{1}_{H\times W}$. Evidence refers to the metric collected from each pixel that supports its classification as either a changed or unchanged pixel. Uncertainty is inversely proportional to the total amount of evidence, meaning that maximum uncertainty score $U = \mathbf{1}_{H\times W}$ is achieved when evidence scores for changed and unchanged pixels are both zero (i.e., $E_c = E_{nc} = \mathbf{0}_{H\times W}$). The probabilities $P = [P_c, P_{nc}]$ can be calculated as follows:

$$\left[P_c,\ P_{nc}\right] = \frac{\left[\boldsymbol{\alpha}_c,\ \boldsymbol{\alpha}_{nc}\right]}{S} \tag{26}$$

where $\boldsymbol{\alpha} = \left[\boldsymbol{\alpha}_c, \boldsymbol{\alpha}_{nc}\right] = \left[E_c, E_{nc}\right] + \mathbf{1}_{H\times W}$. The confidence $\boldsymbol{\alpha}$ enables simultaneous probabilistic density modeling of both CD probabilities and uncertainty. The probability density function is defined as follows:

$$D(P|\boldsymbol{\alpha}) = \begin{cases} \dfrac{1}{B(\boldsymbol{\alpha})}\displaystyle\prod_{k=1}^{K} P^{\alpha_k - 1}, & \text{for } p_{i,j} \in S_K \\ 0, & \text{otherwise} \end{cases} \tag{27}$$

where, $B(\cdot)$ denotes the $K$- dimensional multivariate Beta function, and $S_K$ represents the $K$- dimensional simplex. For binary CD tasks, $K = 2$

$$S_K = \left\{P|\sum_{k=1}^{K} p_{i,j}^k = 1,\ \text{and}\ \forall k,\ 0 \le p_{i,j}^k \le 1\right\}. \tag{28}$$

Furthermore, we identify low-uncertainty pixels (denoted as $P^+$) using the following approach:

$$P^+ = \text{softmax}\left\{P \odot (1 - U)\right\} \odot f\left(\left[F_{u_1} \copyright F_{u_2},\ F_p\right]\right). \tag{29}$$

Here, this operation is equivalent to weighting decoupled features using high-certainty predictions, and the final prediction result is given as follows:

$$P^L = \beta P^+ + (1 - \beta)\, P. \tag{30}$$

To learn the uncertainty map, we model the probability density function of $S$ using a Dirichlet distribution [43] and construct the following evidence loss function:

$$\begin{aligned}\mathcal{L}_E &= \int\left[\sum_{k=1}^{K} Y(k)\log(S_K)\right] D(S|\boldsymbol{\alpha})\, dS \\ &= \sum_{k=1}^{K} Y(k)\left(\psi(S_K) - \psi(\boldsymbol{\alpha})\right)\end{aligned} \tag{31}$$

where $\psi(\cdot)$ denotes the *digamma* function. This loss function can drive the model to reduce prediction uncertainty through evidence accumulation and enhance model adaptation in various scenarios (e.g., varying illumination conditions).

Meanwhile, this article optimizes network parameters by minimizing cross-entropy

$$\mathcal{L}_{BCE} = \frac{1}{H\cdot W}\sum_{h=1,w=1}^{H,W} l_{bce}\left(P_{hw}^L,\ G\ T_{hw}\right). \tag{32}$$

The overall objective loss function $\mathcal{L}_{Total}$ for our proposed network is defined as follows:

$$\mathcal{L}_{Total} = \mathcal{L}_{BCE} + \lambda_1\cdot\mathcal{L}_{DCL} + \lambda_2\cdot\mathcal{L}_E + \lambda_3\cdot\sum_{i=1}^{2}\mathcal{L}_{RE}^{T_i}. \tag{33}$$

## IV. Experimental Setup

### A. Datasets and Evaluation Metrics

To validate the effectiveness of UAFDcd, we conduct comprehensive experiments on three representative RSCD datasets (i.e., LEVIR-CD, CDD, and SECOND). Each dataset consists of paired RS images captured at different times for the same geographic region, along with corresponding CD ground truth labels. For objective quantitative evaluation, we

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adopt five evaluation metrics—precision (P), recall (Re), $F1$-score, overall accuracy (OA), and mean intersection over union (mIoU)—to comprehensively assess model performance

$$P = \frac{TP}{TP + FP} \tag{34}$$

$$Re = \frac{TP}{TP + FN} \tag{35}$$

$$F1 = \frac{2 \times P \times Re}{P + Re} \tag{36}$$

$$OA = \frac{TP + TN}{TP + TN + FP + FN} \tag{37}$$

$$mIoU = \frac{1}{2}\left(\frac{TP}{TP + FP + FN} + \frac{TN}{TN + FN + FP}\right) \tag{38}$$

where $TP$, $FP$, $TN$, and $FN$ denote the counts of True Positives, False Positives, True Negatives, and False Negatives, respectively.

### B. Experimental Parameters

The UAFDcd is implemented based on PyTorch and trained as well as tested on a single NVIDIA RTX 4090 GPU. Additionally, the AdamW optimizer is adopted with the momentum set to 0.9, weight decay set to 5e-3, and initial learning rate set to 0.001. The learning rate of the optimizer is dynamically adjusted, which is reduced by a factor of 0.9 every 20 training epochs. The encoder $f_{E_i}$ is designed with a sequence of layers as follows: Conv2d (3, 32, 3, 2)–BN (32)–ReLU–Conv2d (32, 64, 3, 2)–BN (64)–ReLU–Unet(64,64), where BN is the batch normalization layer. The reconstructor $f_{R_i}$ is composed of ConvTranspose2d (128, 64, 3, 2)–BN (64)–ReLU–ConvTranspose2d (64, 32, 3, 2)–BN (32)–ReLU–Conv2d (32, 3, 3, 1)–Tanh.

## V. Experimental Results

### A. Comparison Methods

To validate the effectiveness of UAFDcd on dual-time RS images, we compare it with several state-of-the-art CD methods, including ICIFNet [44], DMINet [45], USSFC-Net [46], ChangeMamba [47], SEIFNet [48], Convformer-CD [49], STRobustNet [50], and DPCA-CD [51]. For fairness, all comparison methods are implemented using the authors' published code, and their parameters are set according to the original literature. The parameter details of other models are as follows.

1) *ICIFNet:* It utilizes the stochastic gradient descent algorithm to update the network parameters. The batch size, the learning rate, the momentum, the weight decay, and the number of training epochs are set to 8, 1e-2, 0.99, 5e-4, and 200, respectively.
2) *DMINet:* This model is trained with the stochastic gradient descent algorithm using the batch size of 8, the initial learning rate of 1e-2, the momentum of 0.99, the weight decay of 5e-4, and the training epochs (max_epochs) of 200.
3) *USSFC-Net:* During the training process, it applies Kaiming initialization from scratch and uses the Adam optimization algorithm to optimize their model, setting the momentum to 0.99. The weight decay is set to 0.0005. The batch size is 32, and the initial learning rate is 0.0001.
4) *ChangeMamba:* The multitemporal image pairs and associated labels are cropped to 256 × 256 pixels for input to the network, and then the data with the original size are inferred using the trained networks on the test set. During training, they optimize the network using the AdamW optimizer with a learning rate of 1e-4 and a weight decay of 5e-3. The batch size is set to 16.
5) *SEIFNet:* During this experiment, they train the model using the PyTorch framework. The AdamW optimizer is employed with a batch size of 8 and implements linear learning rate decay. The initial learning rate is set to 0.0001, while the momentum and weight decay are set to 0.99 and 0.01, respectively. The model is trained for a total of 200 epochs.
6) *Convformer-CD:* In this experimental trial, they employ PyTorch as the deep learning framework. During the training phase, they use several common data augmentation techniques, including flipping, rescaling, cropping, and Gaussian blurring, all of which are applied during the training phase. Since the Swin Transformer backbone is trained from scratch without using any pretrained models, they trained more epochs compared to other models, with a total of 400 training epochs (max_epoch). Their model uses the Adam algorithm for optimization. The batch size is 32, the initial learning rate is 2e-3, and the learning rate decreases following the cosine schedule.
7) *STRobustNet:* Their model is implemented on PyTorch, and data augmentation is applied to the input image patches, including random flip, random rescale, random crop, Gaussian blur, and random color jitter. They use stochastic gradient descent with a momentum of 0.9 and a weight decay of 0.0005 to optimize the model.
8) *DPCA-CD:* The NVIDIA 3090 24 GB GPU is used as the model training platform; the batch size during training is set to 10, the optimizer is Adam, and the learning rate is 0.001.

Table I presents the quantitative metrics of different models across three datasets. Our model performs strongly on comprehensive metrics (i.e., $F1$-score, OA, and mIoU). For instance, it achieves an $F1$-score of 89.70 on LEVIR. This may be attributed to mechanisms like feature decoupling, which improve semantic boundary clarity and enable precise localization for change regions. However, our model does not perform satisfactorily in terms of metric OA on the SECOND dataset. On the one hand, UAFDcd achieves a relatively high recall rate; on the other hand, the change targets in the SECOND dataset are more concealed and difficult to detect. As a result, our model tends to be aggressive in identifying change regions, leading to a relatively high number of false positives. Nevertheless, our model still achieves the best overall performance (i.e., $F1$-score).

To facilitate intuitive comparison, this article provides a visual comparison of experimental results for various algorithms, as shown in Figs. 5–7.

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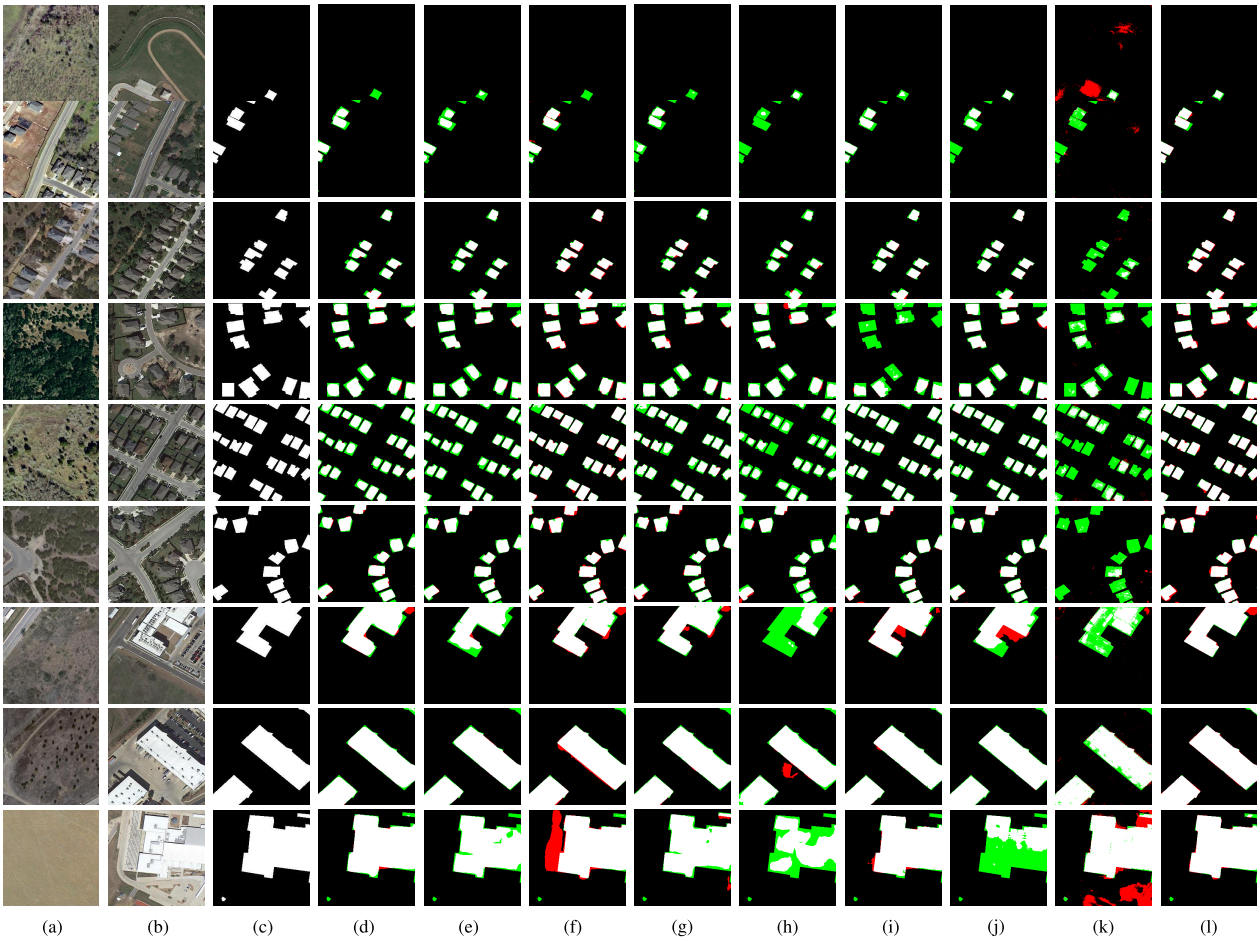



Fig. 5. Detection results on the LEVIR-CD using different methods. (a) T1 Image. (b) T2 Image. (c) Ground Truth. (d) ICIFNet. (e) DMINet. (f) USSFC-Net. (g) ChangeMamba. (h) SEIFNet. (i) Convformer-CD. (j) STRobustNet. (k) DPCA-CD. (l) Ours.

TABLE I
PERFORMANCE COMPARISON OF DIFFERENT METHODS ON VARIOUS DATASETS

| Methods | LEVIR - CD | | | | | CDD | | | | | SECOND | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | P. | Re. | F1 | OA | mIoU | P. | Re. | F1 | OA | mIoU | P. | Re. | F1 | OA | mIoU |
| ICIFNet [44] | 76.65 | 81.25 | 78.88 | 98.30 | 81.69 | 94.71 | 88.56 | 91.53 | 98.07 | 91.11 | **78.21** | 53.96 | 63.86 | 87.79 | 66.61 |
| DMINet [45] | 79.23 | 77.61 | 78.41 | 98.33 | 81.39 | 95.22 | 90.41 | 92.75 | 98.33 | 92.31 | 73.15 | 54.43 | 62.48 | 86.90 | 65.37 |
| USSFC-Net [46] | 94.85 | 73.98 | 83.13 | 98.47 | 84.77 | 93.11 | 83.42 | 88.00 | 97.19 | 87.72 | 74.18 | 58.50 | 65.41 | 87.63 | 67.30 |
| ChangeMamba [47] | 94.16 | 75.81 | 84.01 | 98.53 | 85.44 | **96.20** | 90.38 | 93.20 | 98.37 | 92.71 | 68.89 | 63.95 | 66.33 | 88.56 | 68.05 |
| SEIFNet [48] | **94.87** | 68.24 | 79.38 | 98.20 | 81.97 | 95.92 | 87.21 | 91.36 | 97.73 | 90.90 | 69.29 | 63.19 | 66.10 | 87.04 | 67.26 |
| Convformer-CD [49] | 76.98 | 85.99 | 81.24 | 98.45 | 83.40 | 94.81 | 93.14 | 93.96 | 98.59 | 93.52 | 69.09 | 60.15 | 64.31 | 88.24 | 67.12 |
| STRobustNet [50] | 77.06 | 85.01 | 80.84 | 98.43 | 83.11 | 93.32 | **94.01** | 93.66 | 98.50 | 93.20 | 69.75 | 63.40 | 66.42 | **88.71** | **68.51** |
| DPCA-CD [51] | 81.99 | 81.69 | 81.84 | 98.73 | 83.98 | 94.62 | 84.92 | 89.51 | 97.68 | 89.22 | 72.85 | 59.49 | 65.50 | 87.47 | 67.24 |
| **Ours** | 90.69 | **88.73** | **89.70** | **98.96** | **90.12** | 94.96 | 93.89 | **94.42** | **98.63** | **93.95** | 67.47 | **68.45** | **67.96** | 87.09 | 68.25 |

*All values are reported in percentage (%). Color convention: **best**, 2-nd best.

On the LEVIR-CD dataset, ICIFNet, DMINet, ChangeMamba, and SEIFNet suffer from severe false negatives in small-target regions, while USSFC-Net, Convformer-CD, STRobustNet, and DPCA-CD exhibit high false positive rates in large-target regions. Near complex edges (e.g., densely packed buildings), Convformer-CD, STRobustNet, and DPCA-CD are prone to false positives, whereas DMINet, ChangeMamba, and SEIFNet are prone to false negatives. In contrast, our proposed UAFDcd achieves clear boundaries and low false positive and false negative rates.

On the CDD dataset, ICIFNet, DMINet, USSFC-Net, ChangeMamba, SEIFNet, Convformer-CD, STRobustNet, and DPCA-CD struggle with missed detection of subtle changes and misdetection in edge regions, while ICIFNet, STRobustNet, and DPCA-CD exhibit high false positive rates under seasonal variations. Near complex edges and seasonal changes, most competing methods fail to preserve complete change boundaries and suppress noise. In contrast, our proposed UAFDcd achieves clear boundaries, robust resistance to seasonal variations, and low false positive and false negative rates.

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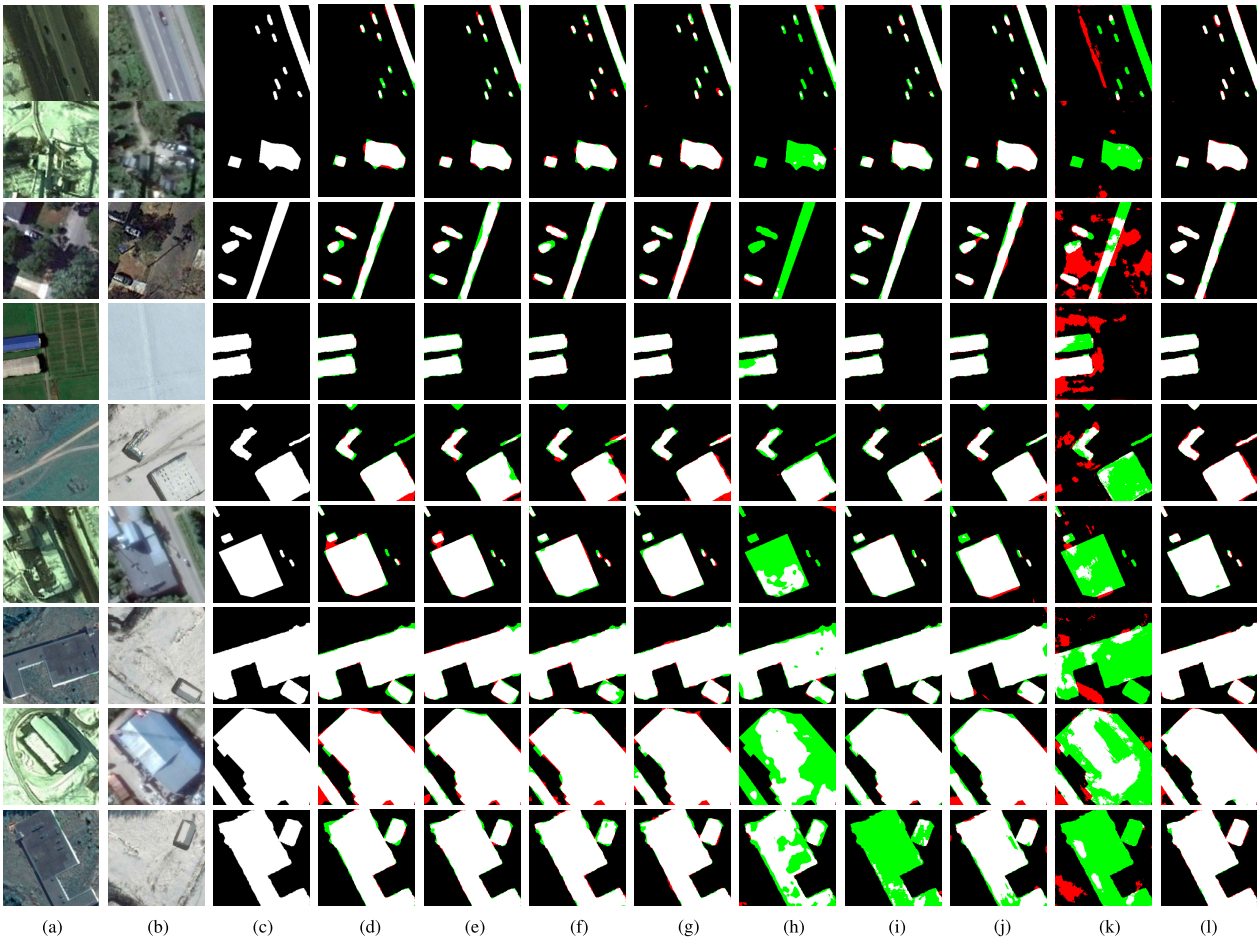



Fig. 6. Detection results on the CDD using different methods. (a) T1 Image. (b) T2 Image. (c) Ground truth. (d) ICIFNet. (e) DMINet. (f) USSFC-Net. (g) ChangeMamba. (h) SEIFNet. (i) Convformer-CD. (j) STRobustNet. (k) DPCA-CD. (l) Ours.

TABLE II
COMPARATIVE EXPERIMENTAL RESULTS

| FD | $\mathcal{L}_{DCL}$ | $\mathcal{L}_E$ | $\mathcal{L}_{RE}^{T_i}$ | P. | Re. | F1 | OA | mIoU |
|---|---|---|---|---|---|---|---|---|
| | ✓ | ✓ | ✓ | 90.64 | 87.34 | 88.96 | 98.90 | 89.48 |
| ✓ | | ✓ | ✓ | **90.89** | 88.23 | 89.54 | 90.89 | 88.23 |
| ✓ | ✓ | | ✓ | 90.37 | 88.44 | 89.40 | 98.93 | 89.85 |
| ✓ | ✓ | ✓ | | 89.34 | **90.00** | 89.67 | 98.94 | 90.09 |
| ✓ | ✓ | ✓ | ✓ | 90.69 | 88.73 | **89.70** | **98.96** | **90.12** |

On the SECOND dataset, ICIFNet, DMINet, USSFC-Net, ChangeMamba, SEIFNet, Convformer-CD, STRobust-Net, and DPCA-CD suffer from severe false positives under perturbed lighting conditions, while ChangeMamba, SEIFNet, and Convformer-CD tend to miss large-scale target changes in complex scenes. DMINet and STRobustNet also struggle with small-target omission and large-target misdetection. In contrast, our proposed UAFDcd outperforms these methods in both target integrity and boundary precision.

## B. Ablation Experiments

*1) Ablation Experiment for Loss Function:* This article validates module effectiveness through ablation experiments for feature decoupler (FD), decoupled contrastive learning $\mathcal{L}_{DCL}$, uncertainty evidence loss $\mathcal{L}_E$, and feature reconstruction loss $\mathcal{L}_{RE}^{T_i}$, etc. As shown in Table II, "✓" indicates that the corresponding module has been adopted. Overall, we remove one module or loss function at a time and observe the changes in model performance. Experimental results show that the model achieves the best performance when all modules are included, indicating that each component makes a positive contribution.

*2) Ablation Experiment for Regularization Coefficients:* It is well known that the insensitivity of experimental results to regularization coefficients indirectly indicates that the model can generalize well. We experimentally tune these regularization hyperparameters on the LEVIR dataset using a control-variable strategy: only one parameter is adjusted at a time while the others are fixed, with changes in the F1 composite metric monitored throughout. For clarity, each set of experimental data corresponding to a given parameter is marked with the same color, as illustrated in Table III. Despite extensive experiments under diverse parameter configurations, the results show no significant fluctuations. Such insensitivity implies that our model does not rely excessively on these hyperparameters. Furthermore, Table I validates that this parameter setting still yields comparatively competitive

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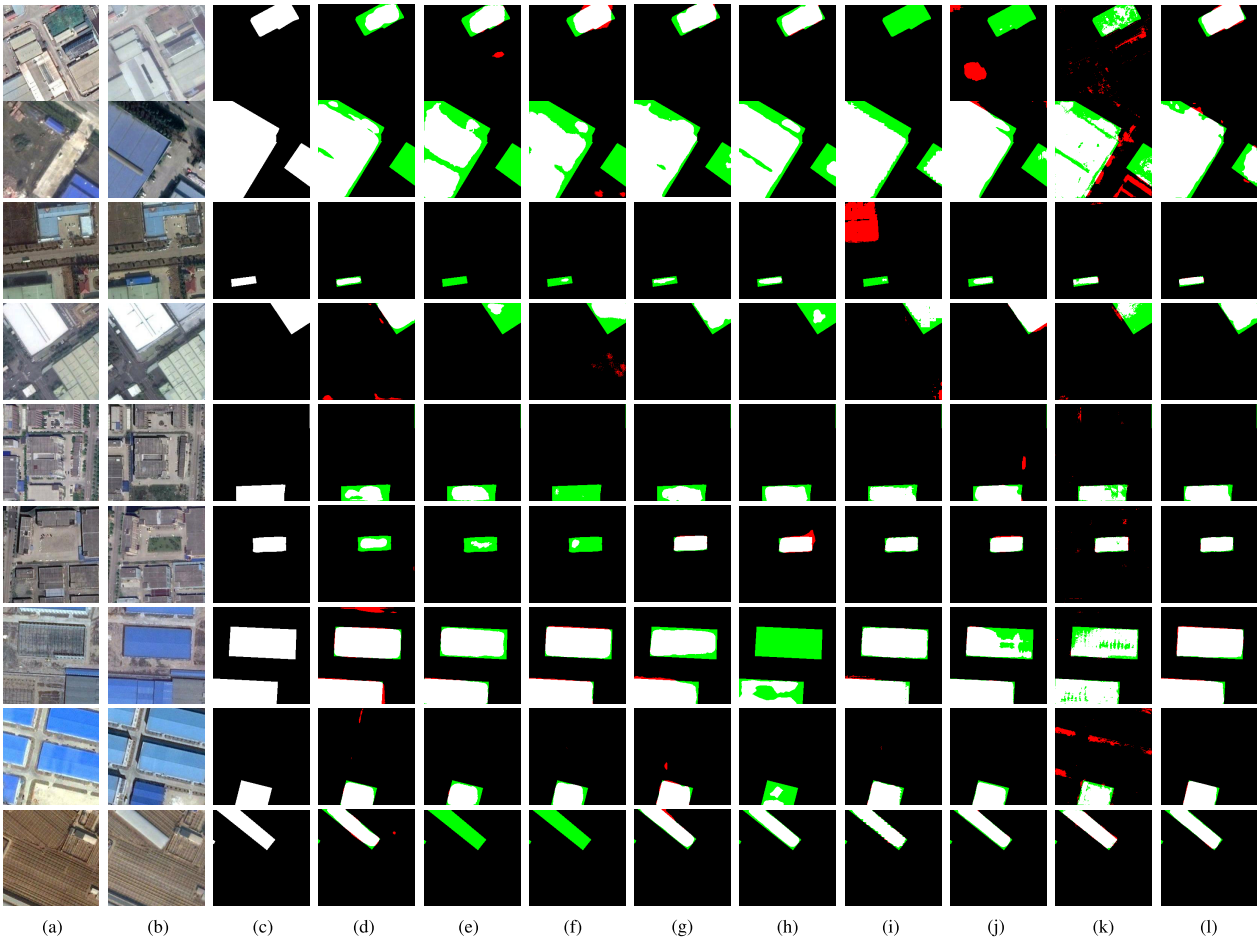



Fig. 7. Detection results on the second using different methods. (a) T1 Image. (b) T2 Image. (c) Ground Truth. (d) ICIFNet. (e) DMINet. (f) USSFC-Net. (g) ChangeMamba. (h) SEIFNet. (i) Convformer-CD. (j) STRobustNet. (k) DPCA-CD. (l) Ours.

TABLE III
INFLUENCE OF REGULARIZATION COEFFICIENT ON LEVIR DATASET

| $\lambda_1$ | $\lambda_2$ | $\lambda_3$ | P. | Re. | F1 | OA | mIoU |
|---|---|---|---|---|---|---|---|
| 1 | 1 | 1 | 90.20 | 88.64 | 89.41 | 98.93 | 89.87 |
| 2 | 1 | 1 | 89.30 | 88.32 | 88.81 | 98.87 | 89.34 |
| 10 | 1 | 1 | 87.51 | **88.84** | 88.17 | 98.79 | 88.79 |
| 0.5 | 1 | 1 | 91.21 | 87.44 | 89.29 | 98.93 | 89.76 |
| 0.01 | 1 | 1 | 90.80 | 85.99 | 88.33 | 98.84 | 88.94 |
| 1 | 2 | 1 | 90.38 | 87.68 | 89.00 | 98.90 | 89.52 |
| 1 | 10 | 1 | 88.52 | 81.06 | 84.63 | 98.50 | 85.90 |
| 1 | 0.5 | 1 | 90.13 | 88.69 | 89.40 | 98.93 | 89.86 |
| 1 | 0.01 | 1 | 90.76 | 88.10 | 89.41 | 98.94 | 89.87 |
| 1 | 1 | 2 | 91.20 | 87.59 | 89.36 | 98.94 | 89.82 |
| 1 | 1 | 10 | 91.41 | 86.51 | 88.89 | 98.90 | 89.43 |
| 1 | 1 | 0.5 | 90.69 | 88.73 | **89.70** | **98.96** | **90.12** |
| 1 | 1 | 0.1 | **91.46** | 87.49 | 89.43 | 98.95 | 89.89 |
| 1 | 1 | 0.01 | 90.67 | 87.19 | 88.90 | 98.89 | 89.43 |

performance on the other datasets, which indirectly confirms the generalization ability of our model.

*3) Ablation Experiment for Prototype Number:* During prototype construction, we select reliable pixel tensors according to the Top-K percentage of both changed and unchanged features. To investigate the effect of prototype number on model performance, we conduct ablation experiments, with the results presented in Table IV. The experimental results demonstrate that an excessive number of prototypes may reduce their representativeness, whereas an insufficient number can make prototypes vulnerable to noise, thus degrading the model CD performance.

TABLE IV
INFLUENCE OF TOP-K VALUE ON LEVIR DATASET

| top-K | P. | Re. | F1 | OA | mIoU |
|---|---|---|---|---|---|
| 5% | 89.79 | **89.09** | 89.44 | 98.93 | 89.89 |
| 10% | **90.69** | 88.73 | **89.70** | **98.96** | **90.12** |
| 15% | 90.58 | 85.39 | 87.91 | 98.80 | 88.59 |

TABLE V
INFLUENCE OF TEMPERATURE COEFFICIENT $\tau$ ON LEVIR DATASET

| $\tau$ | P. | Re. | F1 | OA | mIoU |
|---|---|---|---|---|---|
| 0.10 | 90.56 | 87.24 | 88.87 | 98.89 | 89.40 |
| 0.50 | **90.69** | 88.73 | **89.70** | **98.96** | **90.12** |
| 1.00 | 90.34 | **88.92** | 89.62 | 98.95 | 90.05 |

*4) Ablation Experiment for Temperature Coefficient $\tau$:* Table V presents the experimental tuning results of the temperature coefficient $\tau$. It can be observed that the model achieves

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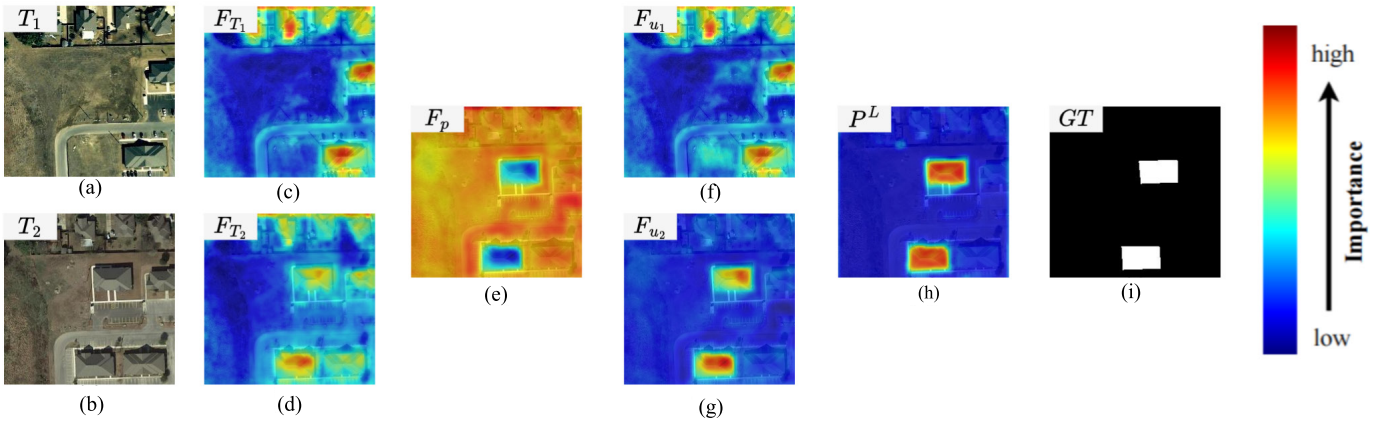



Fig. 8. Visualization of key modules in UAFDcd. (a) Pre-temporal image T1. (b) Post-temporal image T2. (c) and (d) Features $F_{T_i}$ of $T_i$. (e) Decoupled public feature $F_p$. (f) and (g) Decoupled unique feature $F_{u_i}$. (h) Final change probability map. (i) Ground Truth.

TABLE VI
INFLUENCE OF FUSION WEIGHT $\beta$ ON LEVIR DATASET

| $\beta$ | P. | Re. | F1 | OA | mIoU |
|---|---|---|---|---|---|
| 0.30 | **91.24** | 87.86 | 89.52 | 98.95 | 89.96 |
| 0.40 | 90.69 | **88.73** | **89.70** | **98.96** | **90.12** |
| 0.50 | 89.51 | 88.28 | 88.89 | 98.88 | 89.41 |

TABLE VII
OBJECTIVE EVALUATION ABOUT DECOUPLED FEATURE DISCRIMINABILITY

| Methods | SEIFNet | DMINet | USSFC-Net | **Ours** |
|---|---|---|---|---|
| SC | 0.8072 | 0.7052 | 0.7398 | **0.8677** |
| FDR | 3.5164 | 2.4301 | 3.6641 | **10.2375** |

TABLE VIII
CROSS-DATASET GENERALIZATION EVALUATION

| Methods | CDD | | SECOND | |
|---|---|---|---|---|
| | mF1 | mIoU | mF1 | mIoU |
| ICIFNet | 51.51 | 45.49 | 55.57 | 46.47 |
| DMINet | 56.13 | 46.89 | 57.66 | 48.17 |
| USSFC-Net | 53.84 | 44.67 | 51.89 | 43.90 |
| ChangeMamba | 55.72 | 46.42 | 49.48 | 42.63 |
| SEIFNet | 51.08 | 44.95 | 55.49 | 46.72 |
| Convfomer-CD | 47.74 | 44.54 | 50.51 | 44.20 |
| STRobustNet | 50.54 | 45.05 | 49.07 | 43.34 |
| DPCA-CD | 56.46 | 47.65 | 54.16 | 45.42 |
| **Ours** | **58.41** | **47.05** | **58.11** | **48.37** |

the best performance when $\tau = 0.5$. In addition, it is clear that the UAFDcd performance does not fluctuate drastically with variations in $\tau$, indicating that the model is not sensitive to $\tau$.

*5) Ablation Experiment for Fusion Weight $\beta$:* As shown in Table VI, we have experimentally tuned the fusion weight $\beta$. It can be observed that the model achieves the best performance when $\beta = 0.4$. The near-half proportion indicates that reducing uncertainty effectively affects the prediction results.

## C. Visualization

*1) Visualization for Heatmap:* To gain a more intuitive understanding of UAFDcd, we collect two representative sample sets from the LEVIR-CD test dataset to visualize feature maps generated by UAFDcd at different stages, as shown in Fig. 8. Through observation, the effectiveness of feature acquisition of our UAFDcd—especially the relationship between shared and private features—can be summarized as follows.

1) Both the initial features extracted by U-Net and shared/private features obtained after feature separation successfully focus attention on the target area (i.e., buildings).
2) Shared features $F_p$ effectively concentrate attention on shared background regions and associated targets, while private features focus more on their unique targets, particularly $F_{u_2}$.
3) Notably, the attention level of $F_p$ and the initial features of $F_{T_2}$ toward the target buildings should be approximately identical, indicating that their difference features successfully locate the changeable target features; the initial features of $F_{T_1}$ clearly exhibit the higher attention to building targets than $F_p$ which explains why decoupled private feature $F_{u_1}$ can still achieve target locking. In summary, our model not only achieves effective feature separation but also successfully directs feature attention toward the target region across all stages.

*2) Visualization for Feature Separation:* To more intuitively illustrate classification effectiveness, $t$-distributed stochastic neighbor embedding ($t$-SNE) dimensionality reduction is employed for visual analysis. Specifically, it projects high-dimensional embeddings of shared and unique features onto a 2-D plane, thereby enhancing model interpretability. As shown in Fig. 9, yellow and blue dots denote changed and unchanged pixel-level features, respectively. From the visualization results, our UAFDcd exhibits a significantly lower confusion rate between these two class pixels compared to

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TABLE IX
FLOPS AND PARAMS COMPARISON OF DIFFERENT METHODS

| Methods | ICIFNet | DMINet | USSFC-Net | ChangeMamba | SEIFNet | Convformer-CD | STRobustNet | DPCA-CD | **Ours** |
|---|---|---|---|---|---|---|---|---|---|
| FLOPs(G) | 25.36 | 14.55 | 4.86 | 45.74 | 8.37 | 5.14 | 13.19 | 3.1 | **12.08** |
| Params(M) | 23.82 | 6.24 | 1.52 | 17.13 | 27.91 | 37.72 | 5.23 | 3.1 | **8.32** |

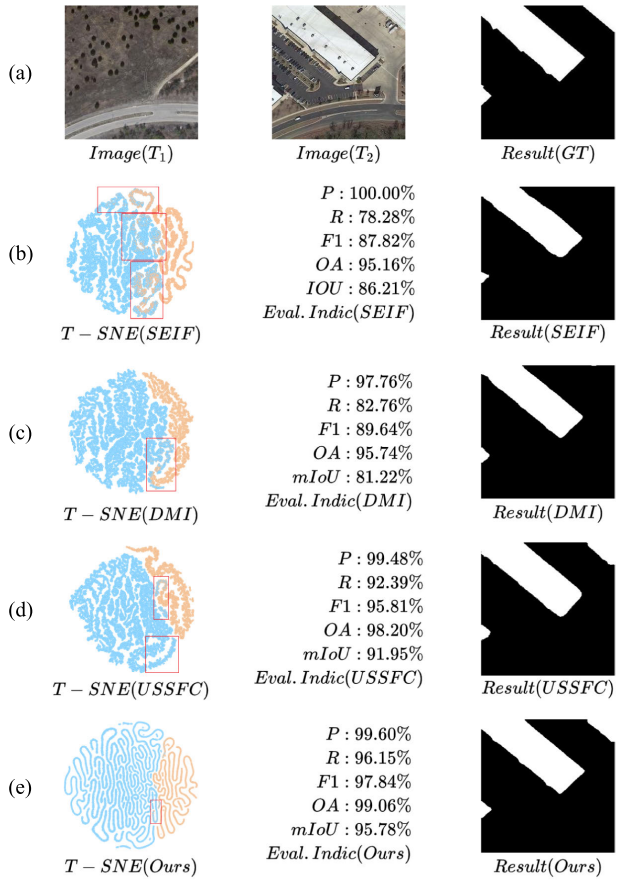



Fig. 9. Some visualization examples produced by typical deep models. (a) Bi-Temporal RS images and GT. (b) Results of the SEIFNet. (c) Results of the DMINet. (d) Results of the USSFC-Net. (e) Results of our UAFDcd.

the others. This indicates that our model enhances the clarity of semantic boundaries, thereby achieving superior CD performance. Objectively, UAFDcd outperforms all comparative models across comprehensive evaluation metrics, including $F1$-score, OA, and mIoU, which is consistent with $t$-SNE visualization results. Regarding prediction results, these key differences are highlighted in red boxes.

To evaluate feature separability, we adopt the Silhouette coefficient (SC) and Fisher discriminant ratio (FDR) to measure the discriminability of decoupled feature representations. As shown in Table VII, both metrics of our model achieve the best performance, indicating that the features of the same class are compact and features derived from different classes are well separated. This is fully consistent with the results of intuitive visualization.

In summary, UAFDcd discriminates between changed and unchanged pixels through feature decoupling, which not only

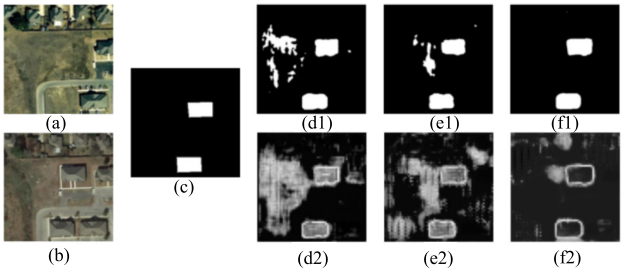



Fig. 10. Analysis of the influence of uncertainty on prediction results. (a) Pre-temporal image T1. (b) Post-temporal image T2. (c) GT. (d1) Uncertainty map at the 10th epoch. (d2) Prediction result at the 10th epoch. (e1) Uncertainty map at the 30th epoch. (e2) Prediction result at the 30th epoch. (f1) Uncertainty map at the 90th epoch. (f2) Prediction result at the 90th epoch.

sharpens semantic boundaries but also improves the interpretability of CD results.

*3) Visualization for Uncertainty Impact:* From the uncertainty maps and prediction results at different epochs, it can be easily observed that regions with high uncertainty usually correspond to large prediction errors, as shown in Fig. 10. In addition, regions with high uncertainty in the final prediction results are usually concentrated at the boundaries of changed areas. This is consistent with the difficulty humans face in that boundaries are difficult to determine when identifying changed areas.

## VI. LIMITATION

Cross-dataset generalization is always one of the important indicators for evaluating deep learning models. As shown in Table VIII, we use the weights trained on the LEVIR dataset to evaluate the CDD and SECOND test sets. The experimental results demonstrate that although the proposed model still has advantages, all performance metrics drop significantly. Although our model is designed with causal feature decoupling to eliminate part of spurious correlation features, the domain distribution shifts caused by complex background interference, seasonal differences, and imaging discrepancies in diverse cross-dataset scenarios still exceed the constraint capability of causal decoupling, which limits the generalization performance of our method. Therefore, cross-dataset generalization remains one of the important research directions for deep learning in the future.

Table IX reports the model complexity. Experiments show that although the parameter number and computational complexity of our model UAFDcd are not the most lightweight, they are still acceptable. To enable its application in more lightweight scenarios, optimizing the model code will be one of the key focuses of our future work.

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## VII. Conclusion

This article proposes the deep network UAFDcd for RSCD based on feature disentanglement to explicitly model the essential features that give rise to changes. First, the nonlinear feature separator is designed to directly obtain the shared features of bi-temporal images as well as the unique features of each temporal image. Then, decouple contrastive learning on prototypes of shared and unique disentangled features is developed to endow the disentangled feature space with both compactness and separability. Finally, an evidence-aware uncertainty estimation module and evidence loss are proposed to enhance the determinacy of disentangled features and model cognitive ability. Experiments demonstrate that UAFDcd achieves promising performance in both comprehensive evaluation metrics and qualitative comparisons on three benchmark datasets, verifying its strong adaptability to CD tasks for diverse target types. However, this model still has room for improvement.

1) The current framework has not yet achieved an optimal trade-off between detection performance and model size, and further optimization is required to improve computational efficiency while maintaining accuracy.
2) Refine the network architecture by drawing inspiration from the human brain. This is intended to mimic the prioritized perception of salient changes in the biological visual system, so as to further alleviate the current performance bottlenecks in RSCD.

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